

Course: 65CS037 – Machine Learning

Task - Regression

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Course: 5CS037/HJ1: Concepts and Technologies of AI

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Date: 31 January 2026

Prediction of Carbon Dioxide Emissions from Existing Buildings Using Regression Models

Abstract

Purpose: The main goal of this project is to predict carbon dioxide emissions (Metric Ton CO₂e) from buildings using regression techniques.

Dataset: The dataset is “Existing Buildings Energy & Water Efficiency (EBEWE) Program” from Los Angeles Open Data Portal. It has about 40,284 useful records after cleaning and 48 features after preprocessing. It connects to United Nations Sustainable Development Goal 11 (Sustainable Cities and Communities) and Goal 13 (Climate Action) because it helps understand and reduce building emissions in cities.

Approach: We did Exploratory Data Analysis (EDA), built a Neural Network, two classical models (Ridge Regression and Random Forest), did hyperparameter tuning, selected important features, and compared all models.

Key Results: The Neural Network gave the best result with RMSE 67.4 and R^2 0.9630. Random Forest also performed well (R^2 around 0.89–0.90). Feature selection showed that only two features (floor area and site EUI) are very important.

Conclusion: The models worked well to predict emissions. Neural Network was strongest, but Random Forest is easier to understand and fast

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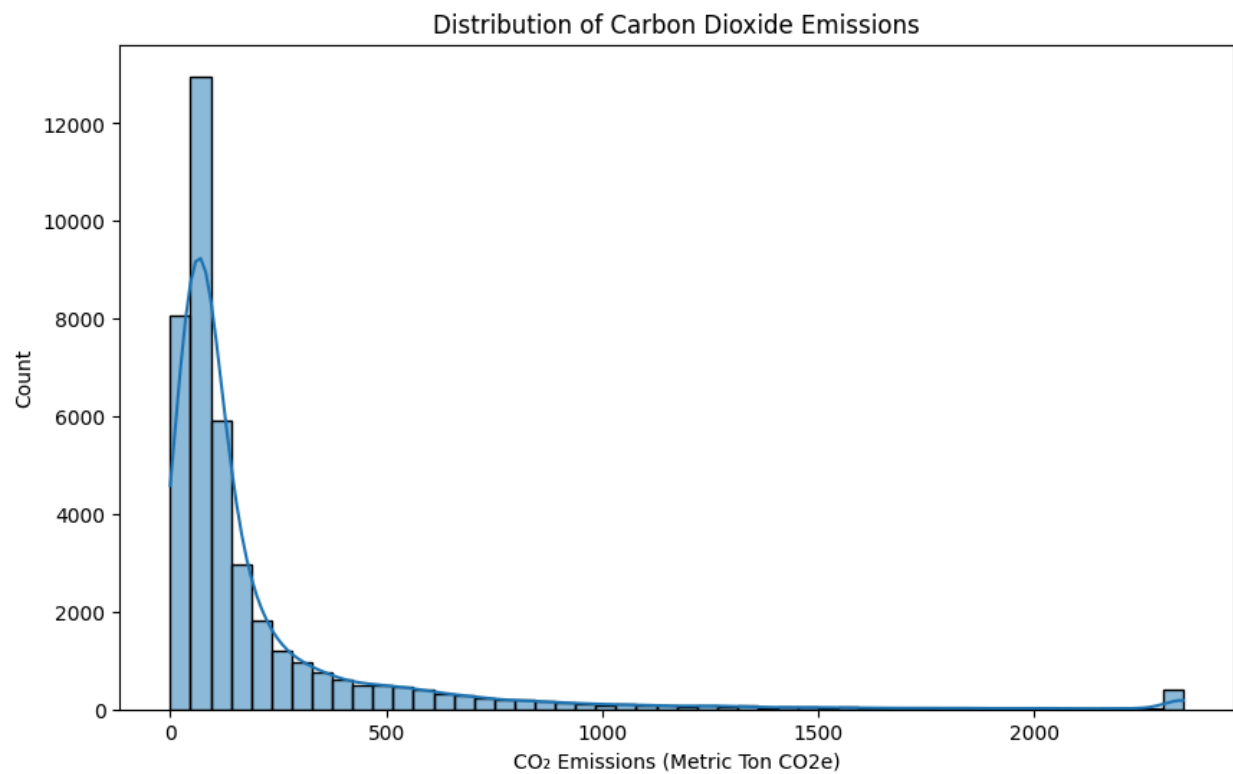


Figure 1: Distribution of CO₂ Emissions (from EDA)



Figure 2: CO₂ Emissions vs Floor Area scatter plot

1. Introduction

1.1 Problem Statement

The aim of this project is to predict carbon dioxide emissions from buildings using machine learning regression models. Accurate prediction helps city planners and governments reduce emissions and make buildings more sustainable.

1.2 Dataset

The dataset is “Existing Buildings Energy & Water Efficiency (EBEWE) Program” from data.lacity.org. It was created by Los Angeles Department of Building and Safety (LADBS) starting from 2016–2017. The data has building information like floor area, year built, property type, energy use intensity (EUI), and CO₂ emissions. After cleaning, we used 40,284 rows.

This dataset supports SDG 11 (Sustainable Cities) and SDG 13 (Climate Action) because it helps track and lower building emissions in urban areas.

1.3 Objective

To build and compare regression models (Neural Network, Ridge Regression, Random Forest) to predict CO₂ emissions accurately using building features.

2. Methodology

2.1 Data Preprocessing

- Removed invalid rows (very small area, impossible age, extreme EUI).
- Capped building age at 250 years and site EUI at 1000.
- Filled missing values with median.
- Scaled numeric features and one-hot encoded categorical features (property type, building category).

2.2 Exploratory Data Analysis (EDA)

We used histograms, scatter plots, box plots, and correlation heatmaps.

Key insights:

- CO₂ emissions are right-skewed (most buildings low, few very high).
- Floor area and site EUI have strong positive relation with emissions.
- Multifamily Housing is the most common property type.
- Older buildings sometimes have higher emissions.

2.3 Model Building

Neural Network (MLP):

- Architecture: 128 → 64 → 32 → 1 neurons
- Activation: ReLU
- Loss: Huber Loss (robust for outliers)
- Optimizer: Adam (lr=0.0005)
- Dropout: 0.3 and 0.2
- Trained for 150 epochs with early stopping and gradient clipping

Two Classical Models:

1. Ridge Regression (improved Linear Regression with regularization)
2. Random Forest Regressor

Data was split 80% train, 20% test.

2.4 Model Evaluation

We used these metrics:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R² Score

2.5 Hyper-parameter Optimization

Ridge Regression:

- Used GridSearchCV (5-fold)
- Tuned alpha: 0.01, 0.1, 1.0, 10, 100, 1000

- Best alpha: 1.0
- Best CV RMSE: 146.0

Random Forest:

- Used RandomizedSearchCV (partial run due to time)
- Tuned n_estimators, max_depth, min_samples_split, etc.
- Used good values for final model (n_estimators=300, max_depth=20, etc.)

2.6 Feature Selection

We used embedded method – Random Forest feature importance.

Selected features with importance > mean.

Only **2 features** were selected:

1. GROSS BUILDING FLOOR AREA (ft²) – 65.9% importance
2. SITE ENERGY USE INTENSITY (EUI) (kBtu/ft²) – 33.3% importance

Justification: These two features explain almost all the prediction power. Emissions mainly depend on building size and how much energy it uses per square foot. Other features (age, property type) add very little after these two. This makes the model simple, fast, and easy to understand.

3. Results and Conclusion

3.1 Key Findings

- Neural Network gave best performance: RMSE = 67.4, $R^2 = 0.9630$
- Random Forest also very good (R^2 around 0.89–0.90)
- Ridge Regression good but weaker than others
- Using only 2 features still gave strong results

3.2 Final Model

Neural Network is the best model in terms of accuracy. But Random Forest with 2 features is also very good and easier to explain.

3.3 Challenges

- Some extreme values caused overflow in log-target model
- Random Forest tuning took long time
- Many one-hot features made model complex

3.4 Future Work

- Try more advanced models like XGBoost or LightGBM
- Add more features (weather data, location)
- Use deep learning with better architecture
- Test on new cities' data

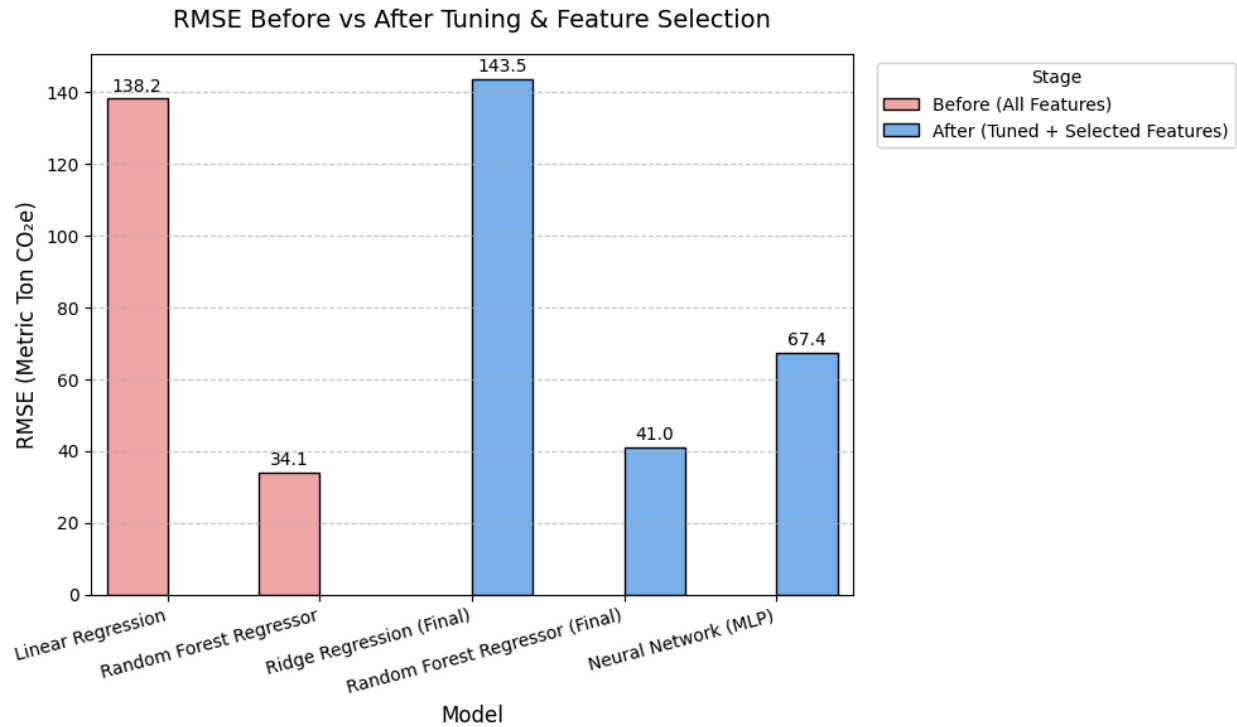
4. Discussion

4.1 Model Performance

Neural Network had highest R^2 (0.9630) and lowest RMSE (67.4).

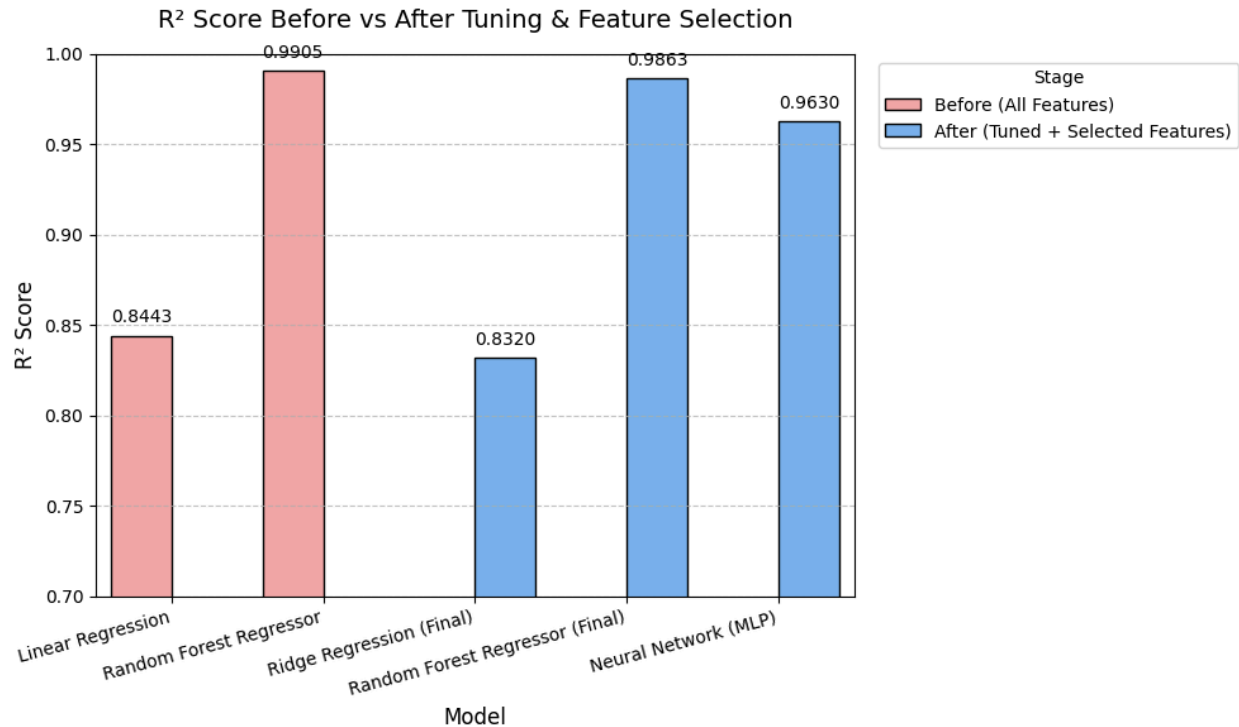
Random Forest was close behind and better than Ridge.

With only 2 features, performance dropped only a little – shows floor area and EUI are most important.



4.2 Impact of Hyperparameter Tuning and Feature Selection

- Tuning alpha in Ridge reduced error a little
- Feature selection to 2 variables made models simpler without big loss in accuracy
- Cross-validation helped choose safe parameters and avoid overfitting



4.3 Interpretation of Results

Floor area and energy intensity are the main drivers of CO₂ emissions. Bigger buildings and less efficient buildings produce much more emissions. Property type and age have small effect after these two.

4.4 Limitations

- Dataset is only from Los Angeles – may not work for other cities
- Many buildings had missing data
- We removed some extreme values

4.5 Suggestions for Future Research

- Use time-series data (multi-year emissions)
- Add climate and energy source features
- Try deep learning with attention mechanism

5. References

1. Los Angeles Open Data Portal: <https://data.lacity.org>
2. Scikit-learn documentation
3. PyTorch documentation
4. United Nations Sustainable Development Goals